Proposed Solution Architecture

Architectural Vision

The proposed solution is a hybrid Clinical Imaging Intelligence Platform designed to orchestrate the end-to-end lifecycle of medical imaging analysis, from DICOM acquisition to AI-assisted clinical decision support.
Rather than relying on a single AI model, the architecture adopts an agentic orchestration approach, where specialized AI agents coordinate multiple analytical services, human validation activities, and enterprise healthcare systems.
The architecture is designed around four fundamental principles:

- **Clinical Safety**: Human oversight remains mandatory for diagnostic decisions.
- **Modularity**: AI capabilities can evolve independently without redesigning the platform.
- **Hybrid Deployment:** Sensitive patient data can remain within hospital environments while leveraging cloud resources when appropriate.
- **Governance by Design**: Security, auditability, explainability, and compliance are built into the architecture.

High-Level Architecture Overview

We define:

A. Execution Domains (NEW PRIMARY LAYERING)

1. **Hospital Edge Domain**
2. **Cloud AI & Data Domain**

This becomes the top-level architectural separation.

B. Capability Layers (inside each domain)

scoped per domain.

Hospital Edge Domain (Clinical Control Plane)

This is now a full stack:

Layer E1 — Clinical Interaction Layer (Edge UI)

- clinician review

- validation tools
- 3D visualization
- annotation tools

Layer E2 — Integration Layer

- PACS connectors
- RIS connectors
- HL7 / DICOM interfaces

Layer E3 — Privacy & Governance Layer

- pseudonymization
- consent enforcement (if needed)
- policy engine

Layer E4 — Edge Workflow Orchestration

- local routing decisions
- caching
- offline resilience

Layer E5 — Secure Sync Layer

- encrypted transfer
- batching
- retry logic

Cloud AI & Data Domain

Layer C1 — AI Processing Layer

- segmentation models
- anomaly detection
- inference endpoints

Layer C2 — 3D Reconstruction Layer

- GPU workloads
- mesh generation
- model conversion

Layer C3 — Data Lake Layer

- S3 storage
- curated datasets
- feedback datasets

Layer C4 — MLOps Layer

- training pipelines
- model registry
- drift detection

Layer C5 — Orchestration Layer

- Step Functions
- EventBridge workflows

Layer C6 — Governance & Audit Layer

- CloudTrail
- lineage tracking
- compliance reporting

Unified Architecture Model

HOSPITAL EDGE DOMAIN

- E1 Clinical UI / Validation
- E2 PACS / RIS / HIS Integration
- E3 Privacy & Pseudonymization
- E4 Local Workflow Engine
- E5 Secure Sync Gateway

CLOUD AWS DOMAIN

- C1 AI Inference (SageMaker)
- C2 GPU Reconstruction (EKS)
- C3 Data Lake (S3)
- C4 MLOps (SageMaker Pipelines)
- C5 Orchestration (Step Functions)
- C6 Governance (Audit, Compliance)

Target Architectural Outcome

The architecture is structured as a dual-domain system composed of a Hospital Edge Domain and a Cloud AI Domain. Each domain contains its own layered architecture. The Edge Domain is responsible for clinical interaction, integration with hospital systems, and privacy enforcement, while the Cloud Domain is responsible for scalable AI computation, data processing, and model lifecycle management.

Responsibility	**Location**
Clinical interaction	Edge
Data privacy enforcement	Edge
AI computation	Cloud
Learning / training	Cloud